

Benchmarking and Performance Analysis of LLM Inferencing Systems

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Simplified Read Kernel



- **Goal:** Model family-workload by kernels that have the sole job of reading KV cache blocks from the HBM
- Each request reads a huge family-shared prefix and a distinct suffix
- Each kernel launch creates its own grid
- All kernels submitted to a single stream

Block-to-Request Mapping



Definitions:

- Blocks per request $= B$
- Total blocks $=$ requests $\times B$

Inside the kernel:

- The request id can be found as: blockIdx.x / B
- The corresponding block in the rqeuest can be found as: blockIdx.x % B

Effect:

- Blocks from different requests execute concurrently
- GPU scheduler interleaves them naturally

Memory Structure




```
[d_mem]
 |-- Family 0
 |     |-- Large Shared Prefix
 |     |-- Request 0 Suffix
 |     |-- Request 1 Suffix
 |     |-- ...
 |-- Family 1
 |     |-- ...
```

Optimizations



CUDA Graph Integration:

- Single kernel launch captured in graph
- Graph instantiated once
- Launched repeatedly (corresponding to multiple decode iterations)

Loop Unrolling:

- Unroll inner loop reading block data
- Each thread reads multiple elements (in a single for loop iteration)

DRAM Accesses' Statistics for vLLM



- **DRAMA:** A DCGMI metric that counts the fraction of cycles the DRAM pipe is active in a unit time interval
- A higher instantaneous value indicates more DRAM activity/utilization
- We sample this metric every 1ms during the kernel execution
- We run two experiments (all in the same batch):
  - Single family with 500 requests
  - 5 families with 100 requests each

DRAM Accesses' Statistics for vLLM



- For the first case, the median DRAM utilization was found to be 0.106, whereas for the second case, it was 0.02, roughly 5x lesser
- Homogeneous batches $\rightarrow$ Higher DRAM utilization
- If we sum over the DRAM utilization samples, we find that the single family case has a sum of 2677.81, whereas the 5 family case has a sum of 3781.25. If we assume that the amount of data transferred is proportional to the sum of DRAM utilizations, we can say that the 5 family case transfers 1.41x more data than the single family case
- This in turn gives us a hint that we are getting higher L2 cache hit rates in the single-family case
- The SM Occupancy was around 0.40 for both the cases

Read Kernel Results (Family Size = 180MB)



For LLama3.1-8B, a shared prefix of length 1300 tokens occupies roughly 180MB.

Metric	Single Family (500 Req)	5 Families (100 Req Each)
Shared Prefix Size (Per Family) (MB)	180	180
Distinct Suffix Size (Per Request) (MB)	3	3
Total HBM Data Needed (MB)	1,760	2,520
Total Data Across All Requests (Including Redundancy) (MB)	95,940	95,940
Execution Time (s)	4.856	4.848
Median DRAM Utilization	0.305	0.294
Median SM Occupancy	0.66	0.66
DRAM Bandwidth (Including Redundancy, assuming 0 L2 reuse) (GB/s)	19.76	19.79
Peak DRAM Bandwidth (GB/s)	960.10	960.10

Read Kernel Results (Family Size = 900MB)



For LLama3.1-8B, a shared prefix of length 6500 tokens occupies roughly 900MB.

Metric	Single Family (500 Req)	5 Families (100 Req Each)
Shared Prefix Size (Per Family) (MB)	900	900
Distinct Suffix Size (Per Request) (MB)	3	3
Total HBM Data Needed (MB)	2,410	5,770
Total Data Across All Requests (Including Redundancy) (MB)	421,000	421,000
Execution Time (s)	21.902	23.086
Median DRAM Utilization	0.391	0.438
Median SM Occupancy	0.66	0.66
DRAM Bandwidth (Including Redundancy, assuming 0 L2 reuse) (GB/s)	19.22	18.01
Peak DRAM Bandwidth (GB/s)	960.10	960.10

Read Kernel Results (Family Size = 1800MB)



For LLama3.1-8B, a shared prefix of length 13000 tokens occupies roughly 1800MB.

Metric	Single Family (500 Req)	5 Families (100 Req Each)
Shared Prefix Size (Per Family) (MB)	1,800	1,800
Distinct Suffix Size (Per Request) (MB)	3	3
Total HBM Data Needed (MB)	3,460	11,010
Total Data Across All Requests (Including Redundancy) (MB)	945,290	945,290
Execution Time (s)	53.733	57.7208
Median DRAM Utilization	0.517	0.557
Median SM Occupancy	0.66	0.66
DRAM Bandwidth (Including Redundancy, assuming 0 L2 reuse) (GB/s)	17.59	16.38
Peak DRAM Bandwidth (GB/s)	960.10	960.10

Read Kernel Results (Family Size = 9000MB)



For LLama3.1-8B, a shared prefix of length 65000 tokens occupies roughly 9000MB.

Metric	Single Family (500 Req)	5 Families (100 Req Each)
Shared Prefix Size (Per Family) (MB)	9,000	9,000
Distinct Suffix Size (Per Request) (MB)	3	3
Total HBM Data Needed (MB)	11,010	47,760
Total Data Across All Requests (Including Redundancy) (MB)	4,720,160	4,720,160
Execution Time (s)	313.94	336.6913
Median DRAM Utilization	0.621	0.688
Median SM Occupancy	0.66	0.66
DRAM Bandwidth (Including Redundancy, assuming 0 L2 reuse) (GB/s)	15.04	14.02
Peak DRAM Bandwidth (GB/s)	960.10	960.10



Thank You